When we introduced line integrals over a curve C starting at point P and ending at point Q, we observed that (in general) the route we took to get from P to Q mattered. When we consider line integrals of vector fields $\mathbf{F}$ over C, it turns out that the choice of route *may* or *may not* matter – it depends on the field you consider.

Definition. Let $\mathbf{F}$ be a vector field defined on an open region D in space, and suppose that for any two points P and Q in D, the line integral

$$\int_C \mathbf{F} \cdot d\mathbf{r}$$

is the same along *any* path C connecting P to Q. Then the line integral is **path independent** in D and the field $\mathbf{F}$ is **conservative** on D.[1]

Notationally, we sometimes write $\displaystyle\int_C \mathbf{F} \cdot d\mathbf{r} = \int_P^Q \mathbf{F} \cdot d\mathbf{r}$

Some quick definitions. A region D is **connected** if it is in 'one piece'. A region D is **simply connected** if it has no holes. A vector field $\mathbf{F}$ is a *gradient vector field* if $\mathbf{F} = \nabla f$ for some f. This function f is called a **potential function** for $\mathbf{F}$.

Theorem. [Fundamental Theorem for Line Integrals] Let C be a smooth curve joining the point P to the point Q in space and parametrized by $\mathbf{r}(t)$, where $a \leqslant t \leqslant b$. Let $\mathbf{F}$ be a gradient vector field given by $\mathbf{F} = \nabla f$. Then

$$\int_C \mathbf{F} \cdot d\mathbf{r} =$$

[1]The term *conservative* comes from physics in reference to fields for which the conservation of energy holds.

As a consequence, of the Fundamental Theorem for Line Integrals, we get the following:

Theorem. Let $\mathbf{F}$ be a vector field whose components are continuous throughout an open connected region D in space. It follows that $\mathbf{F}$ is conservative if and only if $\mathbf{F}$ is a gradient vector field.

Example. Consider the conservative vector field $\mathbf{F}(x, y, z) = \langle yz, xz, xy\rangle$, where $\mathbf{F} = \nabla f$ for $f(x, y, z) = xyz$. Find $\int_C \mathbf{F} \cdot d\mathbf{r}$ on any smooth curve C joining $P(-1, 3, 9)$ and $Q(1, 6, -4)$.

Of course, if a field's line integral is path-independent, then the line integral should yield 0 if we begin and end at the same point. That is, if we are computing a *contour* integral. Specifically:

$\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$ for all closed curves C in D.

$\mathbf{F}$ is conservative on D.

$\mathbf{F} = \nabla f$ for some f on D.

So conservative vector field's are pretty nice! This raises two questions:

- Given any vector field $\mathbf{F}$, how do we determine if its conservative?
- If it is, how do we find its potential function f so that $\mathbf{F} = \nabla f$?

Again, we continue to think of '∇' as the multi-dimensional derivative. Since it encodes partial derivatives, we put $\nabla = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle$, where this operator *acts on functions.*

Definition. Let $\mathbf{F}(x, y, z) = \langle M(x, y, z), N(x, y, z), P(x, y, z)\rangle$ be a vector field. The **curl** of $\mathbf{F}$ is given by

$\text{curl}(\mathbf{F}) = \nabla \times \mathbf{F} =$

Component Test for Conservative Fields Let $\mathbf{F}(x, y, z) = \langle M(x, y, z), N(x, y, z), P(x, y, z)\rangle$ be a vector field on an open simply-connected domain whose component functions have continuous first partial derivatives. Then

$$\mathbf{F} \text{ is conservative} \iff \text{curl}(\mathbf{F}) = \vec{0}$$

In this case, $\mathbf{F} = \nabla f$ for an f satisfying $\dfrac{\partial f}{\partial x} = M$, $\dfrac{\partial f}{\partial y} = N$, and $\dfrac{\partial f}{\partial z} = P$.

Example. Show that $\mathbf{F}(x, y, z) = \langle e^x \cos(y) + yz, xz - e^x \sin(y), xy + z\rangle$ is a conservative vector field, then find a potential function for it.

Example. Show that $\mathbf{F}(x, y, z) = \langle 2x - 3, -z, \cos(z)\rangle$ is not conservative.

Thus, when encountering a conservative vector-field, we can compute its line integral across a smooth curve C from P to Q parametrized by $\mathbf{r}(t)$ in two different ways:

(i) Evaluating
$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \left(\mathbf{F}(\mathbf{r}(t)) \cdot \frac{d\mathbf{r}}{dt}\right) dt$$

(ii) Passing to a potential function f (i.e., $\mathbf{F} = \nabla f$) and using the Fundamental Theorem for Line Integrals to compute
$$\int_C \mathbf{F} \cdot d\mathbf{r} = f(P) - f(Q).$$